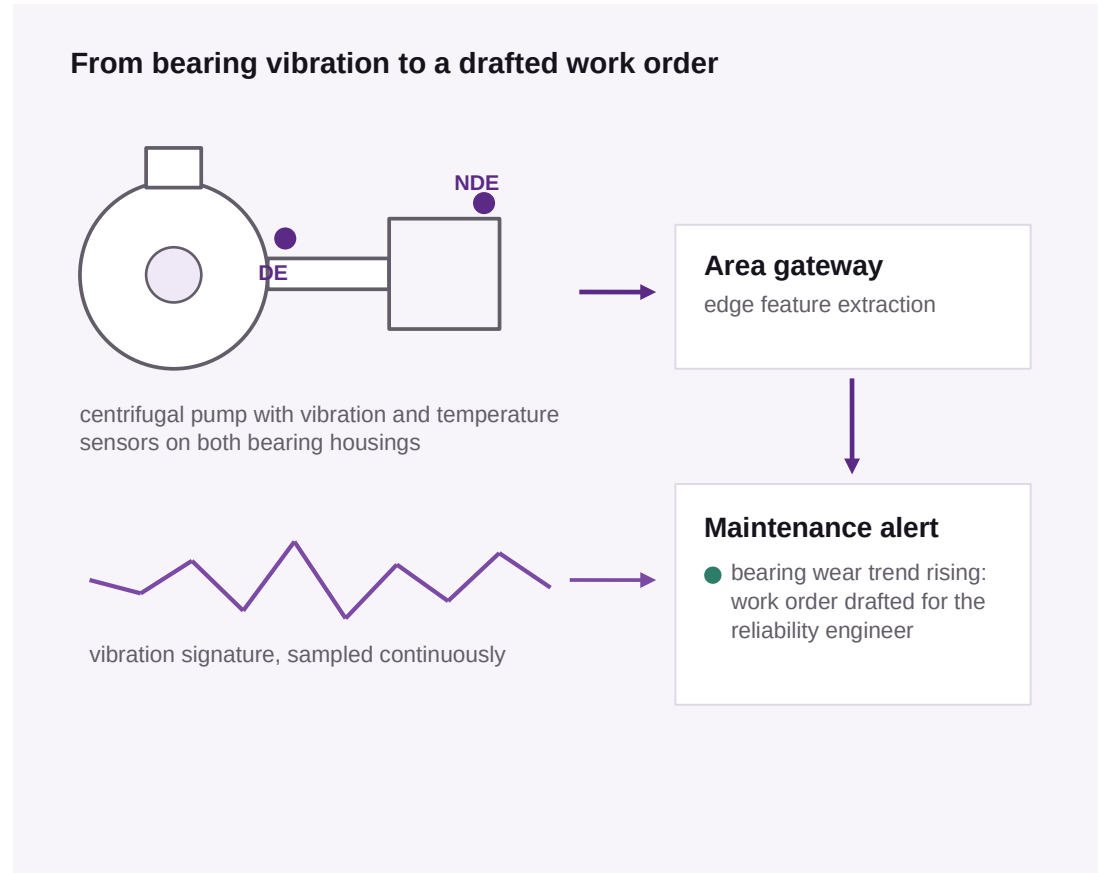


NAADI

Early fault detection for critical pumps, on the plant's own network

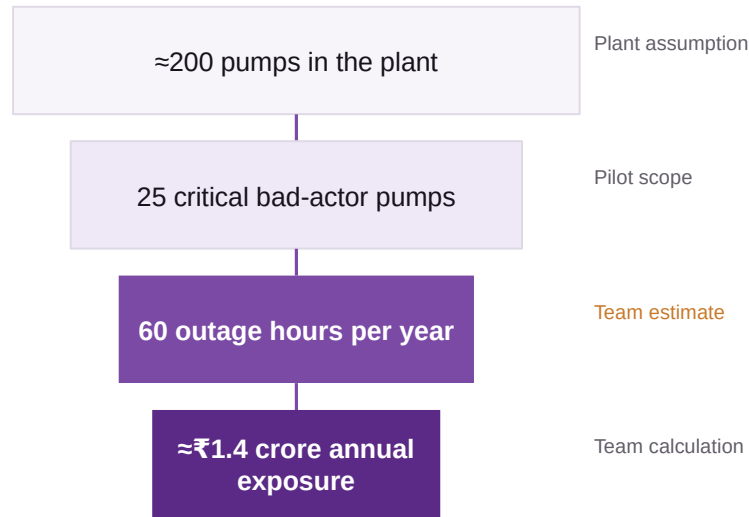
A six-month pilot to test early fault detection across 25 critical pumps in a mid-sized speciality chemicals plant.

Annexure problem statement: AI-driven predictive maintenance to improve asset reliability and maintenance efficiency in the chemicals and petrochemicals sector [1]



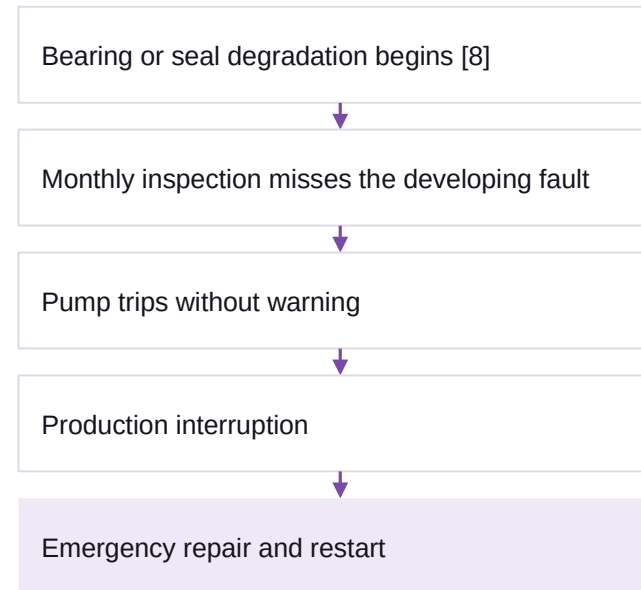
Twenty-five critical pumps can put nearly ₹1.4 crore of annual production value at risk

Where the risk concentrates



How ₹1.4 crore is worked out: 60 h × ₹1.9 lakh per hour = ₹1.14 Cr of lost margin, plus 12 repairs × ₹2.5 lakh = ₹0.30 Cr. Team calculation from the labelled estimates [11].

How a routine fault becomes an outage



89%

of failure modes in the foundational reliability study for RCM showed no predictable wear-out age [7]

Fixed-interval maintenance cannot anticipate faults that arrive at random. Bearing and seal problems develop over days to weeks [8]; that window is where condition data helps.

What NAADI must do: spot developing faults on these 25 pumps early (the target is two weeks of notice) and alert the engineer with a drafted work order.

60 h/yr

unplanned outage from these pumps

Team estimate

₹1.9 lakh/h

contribution margin lost while down

Team calculation

12 events/yr

pump repairs across the 25 units

Team estimate

≈₹1.4 Cr

annual exposure, margin plus repairs

Team calculation

Continuous monitoring works, but proprietary systems remain difficult for mid-sized plants to adopt

Five approaches, six weighted criteria

Approach	Lead time (30)	Cost (20)	Skill need (10)	Data control (10)	Retrofit (10)	MSME fit (20)
Run to failure	none	poor	low	n/a	n/a	poor
Calendar preventive (common in India)	none for random faults	medium	low	n/a	n/a	medium
Monthly offline vibration rounds	up to route interval	medium	high	plant	good	poor
OEM cloud monitoring	days to weeks	high	low	vendor silo	medium	poor
NAADI: on-site continuous stack	days to weeks	low to medium	medium	plant	good	good

Weights in brackets, sum 100. Marks are team judgement based on the documented behaviour of each approach class [2][5].

Weighted score out of 5



The gap is not sensing or algorithms. It is cost, data control and fit for mid-sized plants.

How a score is built: each approach gets a 1-to-5 mark per criterion; the score is the weighted average. NAADI: $(5 \times 30 + 4 \times 20 + 3 \times 10 + 5 \times 10 + 4 \times 10 + 4 \times 20) / 100 = 4.3$.

Design choices



Continuous sensing

no month-long blind windows



Explainable vibration features

auditable by plant engineers



Plant-controlled data

no vendor silo



Retrofit installation

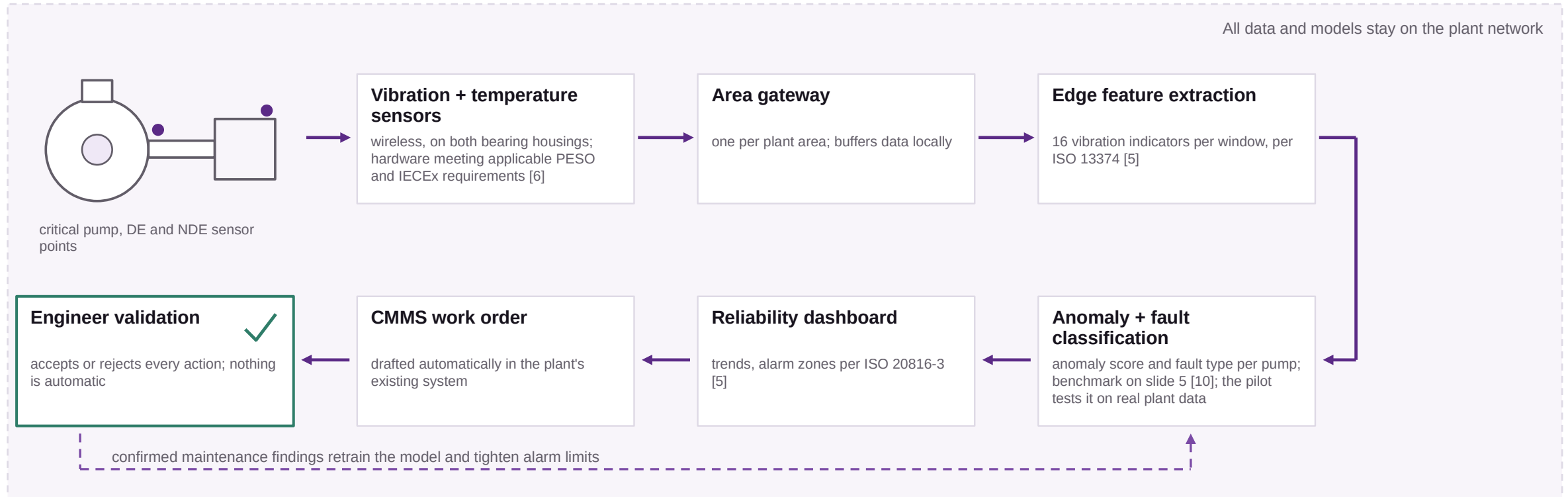
no shutdown, no cabling



Connection to the existing CMMS

alerts become work orders

NAADI links vibration sensing to maintenance action while keeping operational data on site



Open data protocols

OPC UA and standard historian interfaces; nothing proprietary between plant and models



Interpretable features

every model input is a standard vibration indicator an engineer can check

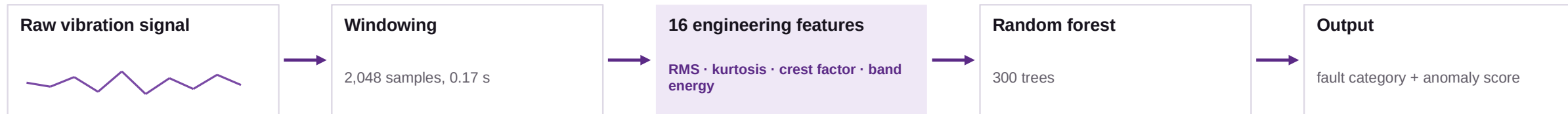


Human-approved action

no maintenance decision changes without the reliability engineer's sign-off

The benchmark result is promising; the plant pilot must now test performance under real operating conditions

Feature pipeline



Prototype benchmark results

97.9%

four-class benchmark accuracy,
unseen 3 hp load [10]

0 of 237

healthy benchmark windows
misclassified [10]

98.9%

grouped cross-validation accuracy [10]

CWRU is a controlled test-rig dataset with seeded faults. These results do not represent proven chemical-plant performance.

Evaluation note: whole recordings and one entire motor load were held out; windows from one recording never appear in both training and test.

Confusion matrix, unseen 3 hp load [10]

	Normal	Inner race	Ball	Outer race
actual				
Normal	237	0	0	0
Inner race	0	216	21	0
Ball	0	0	236	0
Outer race	0	0	3	410

1,123 test windows. Confusions occur only between fault types, never between fault and healthy.

Already established

- Industrial vibration sensors
- Wireless gateways
- OPC UA
- CMMS APIs
- ISO vibration guidance [5]

Still to be validated in the pilot

- Real plant noise
- Variable loading
- Fault lead time
- Sensor placement
- False-alert rate
- Model transfer

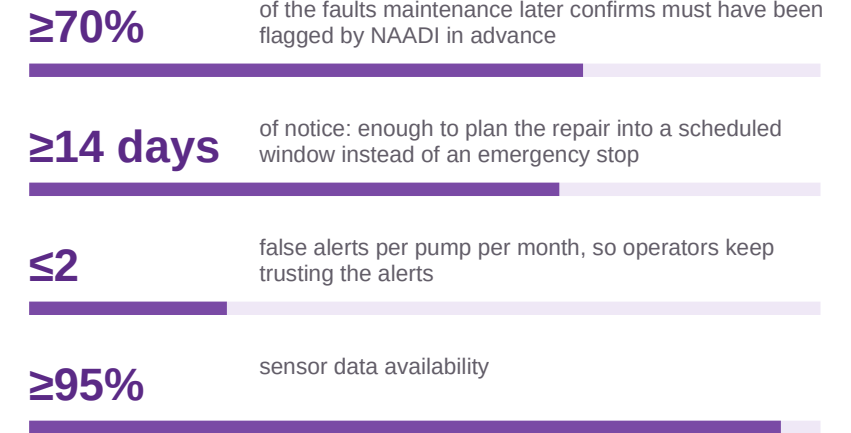
Shadow-mode deployment validates alerts before maintenance teams rely on the system

Six-month pilot plan



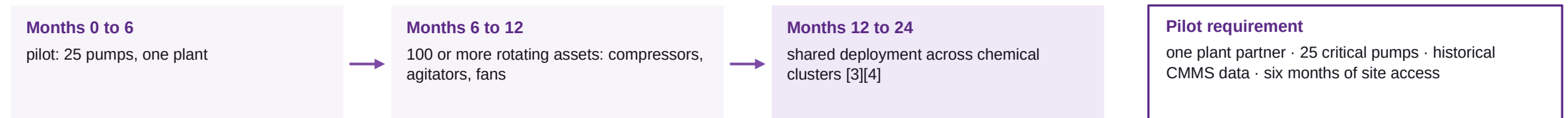
Roles: Reliability engineer: baseline, adjudication, gate · Instrumentation team: install · CMMS administrator: work-order flow · NAADI team: features, models, dashboards

Pilot success criteria



Alerts run in shadow mode for three months and are compared weekly with actual maintenance findings before any alert changes a maintenance decision.

Scale-up after the gate



The base case pays back in 11 months, with a 24-month downside case

₹37 lakh

estimated CapEx, 25 pumps

Team estimate

₹41.4 lakh

annual net benefit, base case

Team calculation

≈11 months

base-case payback

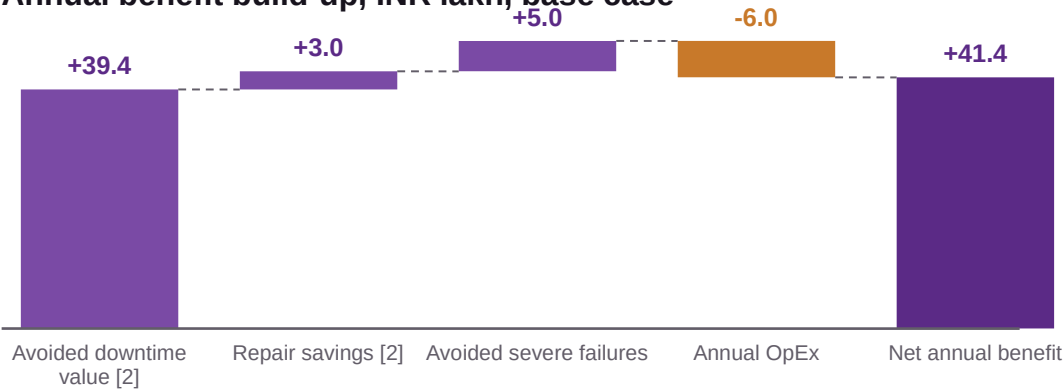
Team calculation

≈24 months

stacked-downside payback

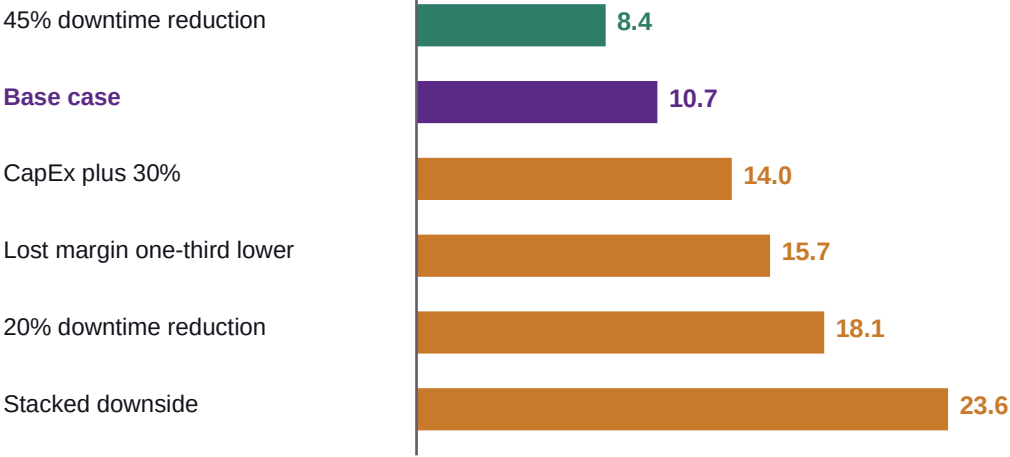
Team calculation

Annual benefit build-up, INR lakh, base case

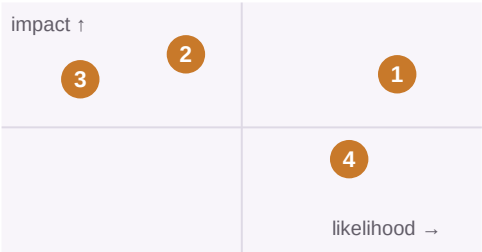


downtime reduction taken at 35%, the low end of the DOE FEMP range [2]

Payback sensitivity, months



Key risks



1 False-alert fatigue

shadow mode first; at most 2 alerts per pump-month at the gate

2 Hazardous-area hardware

hardware meeting applicable PESO and IECEx requirements [6]

3 OT cybersecurity

one-way data flow on a segregated network; no path to the DCS

4 Few early failure examples

anomaly-first models per pump; classifier added as data grows

Environmental gains

- Earlier detection of seal degradation can reduce leak-event risk [8]
- The same condition data can identify cavitation and off-design operation; pumping is close to 20% of world electric-motor demand [9]

Recommendation: proceed with a six-month shadow-mode pilot, subject to plant-data validation and vendor quotations.

Sources, standards and calculation basis

Government and industry guidance

- [1] Department of Chemicals and Petrochemicals (GoI) and FICCI. Youth Innovation Challenge brief, India Chem 2026. 2026. Provided competition document.
- [2] US Department of Energy, FEMP. Operations and Maintenance Best Practices Guide, Release 3.0 (predictive maintenance chapter). 2010. energy.gov
- [3] NITI Aayog. National Strategy for Artificial Intelligence. 2018. And DST, National Mission on Interdisciplinary Cyber-Physical Systems. 2018. dst.gov.in
- [4] Ministry of Heavy Industries. SAMARTH Udyog Bharat 4.0 (Industry 4.0 centres). 2024. samarthudyog-i40.in

Engineering standards

- [5] ISO 20816-3:2022 machine vibration evaluation; ISO 13374-1:2003 condition-monitoring data processing; ISO 17359:2018 condition-monitoring guidelines; ISO 14224:2016 reliability data collection. iso.org
- [6] Petroleum and Explosives Safety Organisation (PESO), DPIIT. Statutory approvals for equipment in hazardous areas. peso.gov.in

Technical literature

- [7] Nowlan, F.S. and Heap, H.F. Reliability-Centered Maintenance. US DoD report AD-A066579. 1978.
- [8] McKee, K. et al. A review of major centrifugal pump failure modes. ICOMS Asset Management Conference. 2011. espace.curtin.edu.au
- [9] Hydraulic Institute, Europump and US DOE. Pump Life Cycle Costs: A Guide to LCC Analysis for Pumping Systems. 2001.

Dataset and team calculations

- [10] Case Western Reserve University Bearing Data Center, seeded-fault vibration dataset. Team model code train_fault_model.py and results metrics.json, 2026. Benchmark results only; not plant performance. engineering.case.edu/bearingdatacenter
- [11] Team economics: NAADI_pilot_economics.xlsx (live formulas and assumption register) and CALCULATIONS.md, 2026. Plant archetype values are team estimates pending pilot data.

Basis of preparation. Estimates and calculations are labelled where they appear. The comparison scores on slide 3 are team judgement. No plant trial, vendor quotation or primary research has been completed yet; the pilot in slide 6 is designed to supply that evidence.